Science - Soil explorers

Year 3 - General practice - Provisional

Name: _____ Date: _____ Class: _____ Score: _____

Original general practice, aligned with the class diary topics: soil, its materials, differences, living things and layers (Unit 4, lessons 1-2; Journal pages 97, 98, 100, 101, 102). Examples are learning tips. The book pages and official assessment criteria were not available. This is not an official school test. Points are for practice only.

Work at your own pace. Circle, write or match as instructed. No soil collection or experiment is needed. An adult may read the instructions aloud.

Part A - Look for the best explanation

2 points

Circle one answer for each question.

1. A soil sample contains tiny rock pieces and old leaf pieces. What does this show?
   a. Soil is only broken rock.
   b. Soil is a mixture of materials.
   c. Soil is only fallen leaves.

2. Some spaces between soil grains contain water. What can other spaces contain?
   a. Air between the grains.
   b. Only more solid grains.
   c. Only bits of old leaves.

3. Two soil samples have the same colour. Which statement is careful?
   a. Their grains must match too.
   b. They must have equal water.
   c. Their grains may still differ.

4. You want to show different layers below the surface. Which drawing helps?
   a. A view of only the surface.
   b. A side view showing bands.
   c. A view with all bands mixed.

5. A root grows down into soil. Which idea does this support?
   a. Roots grow only in air.
   b. Soil is always lifeless.
   c. Soil supports living things.

6. An old leaf breaks into small pieces on the ground. What can happen to those pieces?
    a. They can become part of soil.
    b. They all turn into solid stones.
    c. They stop being plant remains.

7. One sample has large grains. Another has small grains. What is different?
    a. The number of soil layers.
    b. The size of the soil pieces.
    c. The amount of water present.

8. A worm disappears into the soil. Why can it be missing from a surface picture?
    a. All worms stay above ground.
    b. A hidden worm is not alive.
    c. The worm is below the surface.

Part B - Check each statement

2 points

Write T or F. Correct each false statement.

9. ( ) A worm can live inside soil.
   Correct it: _____

10. ( ) Soil is always made of just one material.
    Correct it: _____

11. ( ) Every soil sample has exactly the same grain size.
    Correct it: _____

12. ( ) Layers can look different below the surface.
    Correct it: _____

Part C - Soil words

2 points

Use four words. One word is extra.

Word bank: layers · mixture · roots · air · surface

13. Soil contains different materials. It is a _____ .

14. Some spaces between soil pieces can hold _____ .

15. Plant _____ can grow down into soil.

16. A side view can show different _____ below the surface.

Part D - Match the clue

2 points

Write each left-hand number beside its matching meaning.

1. Mixture
2. Grain size
3. Worm in soil
4. Side cutaway

( ) a. An example of life below the surface

( ) b. More than one kind of material together

( ) c. A view that can show layers underground

( ) d. How large or small the soil pieces are

Part E - Use two clues

2 points

A short answer is enough. You may explain aloud to an adult.

17. A soil model shows only stones. Name two different things you could add to show what soil contains or supports.

18. Two jars have soil of the same colour. Explain why this does not prove the samples are identical. Name one other feature to compare.

Good luck! · Total: 10

Answer key (Gabarito) — Science - Soil explorers

Part A - Look for the best explanation

1. b) Soil is a mixture of materials. — *The sample contains more than one kind of material.*
2. a) Air between the grains. — *Soil spaces can contain air or water.*
3. c) Their grains may still differ. — *Matching colour does not mean that all other features match.*
4. b) A side view showing bands. — *A side cutaway can reveal layers below the surface. Real layers vary.*
5. c) Soil supports living things. — *Plant roots are living parts that can grow into soil.*
6. a) They can become part of soil. — *Remains of living things, such as old leaf pieces, can become part of soil.*
7. b) The size of the soil pieces. — *Grain size describes how large or small the soil pieces are. It does not tell us the layer count or water amount.*
8. c) The worm is below the surface. — *A surface picture may not show living things underground.*

Part B - Check each statement

9. T (true)
10. F (false) — Soil is a mixture, including rock particles and remains of living things.
11. F (false) — Soil samples can have different grain sizes.
12. T (true)

Part C - Soil words

13. mixture
14. air
15. roots
16. layers

Part D - Match the clue

a → 3 · b → 1 · c → 4 · d → 2

Part E - Use two clues

17. **Expected answer (example):** For example: old leaf pieces and a worm. Other accepted examples include plant roots, air spaces, and water in spaces.
    Para o responsável, confira: 0,5 por cada exemplo diferente e cientificamente adequado; máximo 1 ponto; Aceitar resposta oral; não avaliar ortografia; Não exigir todos os elementos nem sugerir que toda amostra tenha todos eles
18. **Expected answer (example):** The same colour is only one matching feature. Grain sizes or visible materials may differ, so I would compare those too.
    Para o responsável, confira: 0,5 por reconhecer que a mesma cor não prova igualdade total; 0,5 por citar tamanho dos grãos ou materiais visíveis para comparar; Aceitar resposta oral e outras formulações equivalentes